The Ephemeris

March 2021
Volume 32 Number 01- The Official Publication of the San Jose Astronomical Association

March 2021 - June 2021 Events
Schedule subject to change due to Covid-19 restrictions.
Please check our website for up to date information.

Board & General Meetings (online)
Saturday 3/27, 4/24, 5/22, 6/26
Board Meetings: 6 -7:30pm
General Meetings: 8:00-9:30pm

Fix-It Day (2-4pm)
Currently on hold due to COVID-19 Shelter in Place

Solar Sunday (1:30-3:30pm)
Sunday 3/15, 4/5, 5/3, 6/7

Intro to the Night Sky Class &
Houge Park 1Q In-Town Star Party
Friday 4/3, 5/1, 5/29, 6/26

Houge Park 3Q In-Town Star Party
Friday 3/13, 4/17, 5/15, 6/12

RCDO Starry Nights Star Party
Currently on hold due to COVID-19 Shelter in Place

Imaging SIG Mtg
Tuesday 3/17, 4/21, 5/19, 6/16

Astro Imaging Clinics
(at Little Uvas OSP)
Currently on hold due to COVID-19 Shelter in Place

Binocular Star Gazing
Saturday (will resume in Summer 2021)

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

INSIDE THIS ISSUE

The Cave Nebula
Photo by SJAA Member: Alan Pham
January 24, 2021
Location: San Jose, CA

SJAA Annual Awards and Board Elections

Credit: Abishek

SJAA announced the results of its board member elections and annual awards at the February 27th board meeting. It was a great meeting with President Ken Miura presenting the awards right after the board meeting ended. The potluck dinner that usually follows this memorable yearly event was cancelled due to COVID-19 related restrictions.

The board of directors presented certificates of appreciation and a gift to two SJAA members in grateful recognition of their continued service and support. Their generous commitment of time, energy and service in their role is immensely appreciated! The board values their skills in leadership, and in their abilities to be an inspiration for other members of the amateur astronomy community.

The recipients of these awards were:

Emi Nikolov - For her outstanding contribution while serving as the secretary of SJAA since 2020 and as the production editor of the Ephemeris since 2017.

Outstanding Service

IN GRATEFUL RECOGNITION OF YOUR CONTINUING SERVICE AND SUPPORT,
WE, THE BOARD OF DIRECTORS OF THE SAN JOSE ASTRONOMICAL ASSOCIATION, HEREBY PRESENT

EMILY NIKOLOV

WITH THIS CERTIFICATE OF APPRECIATION
FOR YOUR GENEROUS COMMITMENT OF TIME, ENERGY,
SERVICE, LEADERSHIP AND INSPIRATION,
WHILE SERVING AS THE SECRETARY OF ASSOCIATION
AND AS THE PRODUCTION EDITOR OF THE EPHEMERIS

San Jose Astronomical Association SJAA

Kenichi Miura
KENICHI MIURA, SJAA PRESIDENT
February 27, 2021

Emi Nikolov—Secretary and Production Editor of the Ephemeris and her certificate of Outstanding Service.

Rashi Girdhar - For his outstanding contribution in co-sponsoring the starry nights program and serving as one of the leaders of the “Armchair Star Party”.

Outstanding Service

IN GRATEFUL RECOGNITION OF YOUR CONTINUING SERVICE AND SUPPORT,
WE, THE BOARD OF DIRECTORS OF THE SAN JOSE ASTRONOMICAL ASSOCIATION, HEREBY PRESENT

RASHI GIRDHAR

WITH THIS CERTIFICATE OF APPRECIATION FOR YOUR
GENEROUS COMMITMENT OF TIME, ENERGY, SERVICE,
LEADERSHIP AND INSPIRATION,
WHILE CO-SPONSORING THE STARRY NIGHT PROGRAM AND
SERVING AS ONE OF THE LEADERS OF “ARMCHAIR STAR PARTY”
PROGRAM

San Jose Astronomical Association SJAA

Kenichi Miura
KENICHI MIURA, SJAA PRESIDENT
February 27, 2021

Rashi Girdhar — Co-Sponsor of the Starry Nights Star Party and one of the leaders of the “Armchair Star Party” and his certificate of Outstanding Service

Results of the board member elections involved the re-election of Vini Carter, Ken Miura, Swami Nigam, Peter Melhus and Sukhada Palav for the positions of Director 1, 3, 5, 7 and 9 respectively.

A heartiest congratulations to all the candidates!! The San Jose Astronomical Association appreciates your efforts and contributions!

In Memoriam Bob Ayers

By: Rob Jaworski

SJAA crew heading down to Bob Ayer's Willow Springs 3000 site in 2013. Bob is on the far right.

Credit: Ed Wong

There was quite a stir caused for several years when the leadership of the SJAA learned that longtime member Bob Ayers was considering a gift of a remote observing site to the organization. Though it never came through, the would-be gift was a parcel of about forty acres northeast of Pinnacles National Park. In the end, Bob and the club determined that it wasn't quite the right fit. However, the proposal caused a fair amount of reflection and introspection which resulted in an affirmation that the SJAA is a welcoming place for newcomers to the hobby and a good home for more experienced observers and imagers.

Bob passed away at home in October 2020.

He left behind a lifelong love of astronomy, science and technology. Graduating from Harvard with a degree in astronomy, Bob went on to work at a series of technology companies throughout his life, including General Electric, Xerox, Digital Equipment Corp (DEC), and Adobe Systems, from which he retired in 2005. He has a dozen US patents to his name.

After retirement, Bob served as a member of the Lowell Observatory advisory board and a member of the executive committee of the board. More recently, Lowell Observatory named asteroid #25154 "Ayers" in appreciation of his contributions. His Robert Martin Ayers Sciences Fund was put in place to encourage astronomy and science research, especially as it relates to education, which lines up exceptionally well with the mission of the SJAA. In fact, the SJAA has received grants from the Ayers Sciences Fund, for which we are truly grateful. He pursued other passions as well, such as mountaineering, scuba diving and photography, all after he retired.

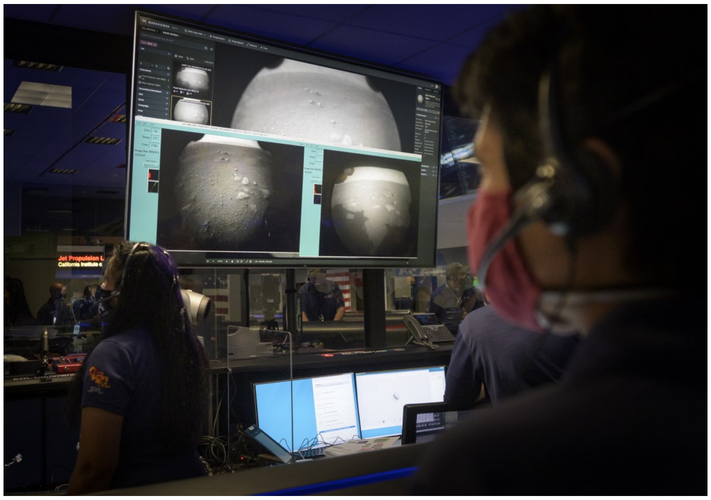
In the end, the SJAA did not take possession of Willow Springs 3000, the acreage in the remote central California hills. Bob sold that place after moving all his gear to another dark sky site he bought, which is about fifty miles west of Flagstaff, Arizona. His goal was to build an observatory there, away from light polluted cities while taking into consideration other lessons he learned from WS3K.

Bob Ayers wasn't a regular at SJAA meetings or in town star parties, nor did he serve on the board or other prominent positions within the organization, as far as we know. Despite that, he did support the club and its mission to bring astronomy and science to the public, educating anyone about the wonders of the universe, and giving kids and grownups alike a place to go to where they can move their spark of interest into a flame of passion for things much bigger, and much smaller, than ourselves. Bob understood that education was the key for living a life worth living.

The board and leaders truly appreciate all that Bob had done for not just the SJAA, but for amateur astronomy and science education in general. We acknowledge and appreciate not just the gifts provided by his Sciences Fund but also the bequest to the SJAA. Thank you, Bob, we will miss you.

Touchdown! NASA's Mars Perseverance Rover Safely Lands on Red Planet

Credits: NASA



Members of NASA's Perseverance Mars rover team watch in mission control as the first images arrive moments after the spacecraft successfully touched down on Mars, Thursday, Feb. 18, 2021, at NASA's Jet Propulsion Laboratory in Pasadena, California. A key objective for Perseverance's mission on Mars is astrobiology, including the search for signs of ancient microbial life. The rover will characterize the planet's geology and past climate, pave the way for human exploration of the Red Planet, and be the first mission to collect and cache Martian rock and regolith.

Credits: NASA

The largest, most advanced rover NASA has sent to another world touched down on Mars Thursday, after a 203-day journey traversing 293 million miles (472 million kilometers). Confirmation of the successful touchdown was announced in mission control at NASA's Jet Propulsion Laboratory in Southern California at 3:55 p.m. EST (12:55 p.m. PST).

Packed with groundbreaking technology, the Mars 2020 mission launched July 30, 2020, from Cape Canaveral Space Force Station in Florida. The Perseverance rover mission marks an ambitious first step in the effort to collect Mars samples and return them to Earth.

"This landing is one of those pivotal moments for NASA, the United States, and space exploration globally – when we know we are on the cusp of discovery and sharpening our pencils, so to speak, to rewrite the textbooks," said acting NASA Administrator Steve Jurczyk. "The Mars 2020 Perseverance mission embodies our nation's spirit of persevering even in the most challenging of situations, inspiring, and advancing science and exploration. The mission itself personifies the human ideal of persevering toward the future and will help us prepare for human exploration of the Red Planet."

About the size of a car, the 2,263-pound (1,026-kilogram) robotic geologist and astrobiologist will undergo several weeks of testing before it begins its two-year science investigation of Mars' Jezero Crater. While the rover will investigate the rock and sediment of Jezero's ancient lakebed and river delta to characterize the region's geology and past climate, a fundamental part of its mission is astrobiology, including the search for signs of ancient microbial life. To that end, the Mars Sample Return campaign, being planned by NASA and ESA (European Space Agency), will allow scientists on Earth to study samples collected by Perseverance to search for definitive signs of past life using instruments too large and complex to send to the Red Planet.

"Because of today's exciting events, the first pristine samples from carefully documented locations on another planet are another step closer to being returned to Earth," said Thomas Zurbuchen, associate administrator for science at NASA. "Perseverance is the first step in bringing back rock and regolith from Mars. We don't know what these pristine samples from Mars will tell us. But what they could tell us is monumental – including that life might have once existed beyond Earth."

Some 28 miles (45 kilometers) wide, Jezero Crater sits on the western edge of Isidis Planitia, a giant impact basin just north of the Martian equator. Scientists have determined that 3.5 billion years ago the crater had its own river delta and was filled with water.

The power system that provides electricity and heat for Perseverance through its exploration of Jezero Crater is a Multi-Mission Radioisotope Thermoelectric Generator, or MMRTG. The U.S. Department of Energy (DOE) provided it to NASA through an ongoing partnership to develop power systems for civil space applications.

Equipped with seven primary science instruments, the most cameras ever sent to Mars, and its exquisitely complex sample caching system – the first of its kind sent into space – Perseverance will scour the Jezero region for fossilized remains of ancient microscopic Martian life, taking samples along the way.

"Perseverance is the most sophisticated robotic geologist ever made, but verifying that microscopic life once existed carries an enormous burden of proof," said Lori Glaze, director of NASA's Planetary Science Division. "While we'll learn a lot with the great instruments we have aboard the rover, it may very well require the far more capable laboratories and instruments back here on Earth to tell us whether our samples carry evidence that Mars once harbored life."

Continued from NASA Mars Perseverance ...

Paving the Way for Human Missions

"Landing on Mars is always an incredibly difficult task and we are proud to continue building on our past success," said JPL Director Michael Watkins. "But, while Perseverance advances that success, this rover is also blazing its own path and daring new challenges in the surface mission. We built the rover not just to land but to find and collect the best scientific samples for return to Earth, and its incredibly complex sampling system and autonomy not only enable that mission, they set the stage for future robotic and crewed missions."

The Mars Entry, Descent, and Landing Instrumentation 2 (MEDLI2) sensor suite collected data about Mars' atmosphere during entry, and the Terrain-Relative Navigation system autonomously guided the spacecraft during final descent. The data from both are expected to help future human missions land on other worlds more safely and with larger payloads.

On the surface of Mars, Perseverance's science instruments will have an opportunity to scientifically shine. Mastcam-Z is a pair of zoomable science cameras on Perseverance's remote sensing mast, or head, that creates high-resolution, color 3D panoramas of the Martian landscape. Also located on the mast, the SuperCam uses a pulsed laser to study the chemistry of rocks and sediment and has its own microphone to help scientists better understand the property of the rocks, including their hardness.

Located on a turret at the end of the rover's robotic arm, the Planetary Instrument for X-ray Lithochemistry (PIXL) and the Scanning Habitable Environments with Raman & Luminescence for Organics & Chemicals (SHERLOC) instruments will work together to collect data on Mars' geology close-up. PIXL will use an X-ray beam and suite of sensors to delve into a rock's elemental chemistry. SHERLOC's ultraviolet laser and spectrometer, along with its Wide Angle Topographic Sensor for Operations and eNgineering (WATSON) imager, will study rock surfaces, mapping out the presence of certain minerals and organic molecules, which are the carbon-based building blocks of life on Earth.

The rover chassis is home to three science instruments, as well. The Radar Imager for Mars' Subsurface Experiment (RIMFAX) is the first ground-penetrating radar on the surface of Mars and will be used to determine how different layers of the Martian surface formed over time. The data could help pave the way for future sensors that hunt for subsurface water ice deposits.

Also with an eye on future Red Planet explorations, the Mars Oxygen In-Situ Resource Utilization Experiment (MOXIE) technology demonstration will attempt to manufacture oxygen out of thin air – the Red Planet's tenuous and mostly carbon dioxide atmosphere.

The rover's Mars Environmental Dynamics Analyzer (MEDA) instrument, which has sensors on the mast and chassis, will provide key information about present-day Mars weather, climate, and dust.

Currently attached to the belly of Perseverance, the diminutive Ingenuity Mars Helicopter is a technology demonstration that will attempt the first powered, controlled flight on another planet.

Project engineers and scientists will now put Perseverance through its paces, testing every instrument, subsystem, and subroutine over the next month or two. Only then will they deploy the helicopter to the surface for the flight test phase. If successful, Ingenuity could add an aerial dimension to exploration of the Red Planet in which such helicopters serve as a scouts or make deliveries for future astronauts away from their base.

Once Ingenuity's test flights are complete, the rover's search for evidence of ancient microbial life will begin in earnest.

"Perseverance is more than a rover, and more than this amazing collection of men and women that built it and got us here," said John McNamee, project manager of the Mars 2020 Perseverance rover mission at JPL. "It is even more than the 10.9 million people who signed up to be part of our mission. This mission is about what humans can achieve when they persevere. We made it this far. Now, watch us go."

More About the Mission

A primary objective for Perseverance's mission on Mars is astrobiology research, including the search for signs of ancient microbial life. The rover will characterize the planet's geology and past climate and be the first mission to collect and cache Martian rock and regolith, paving the way for human exploration of the Red Planet.

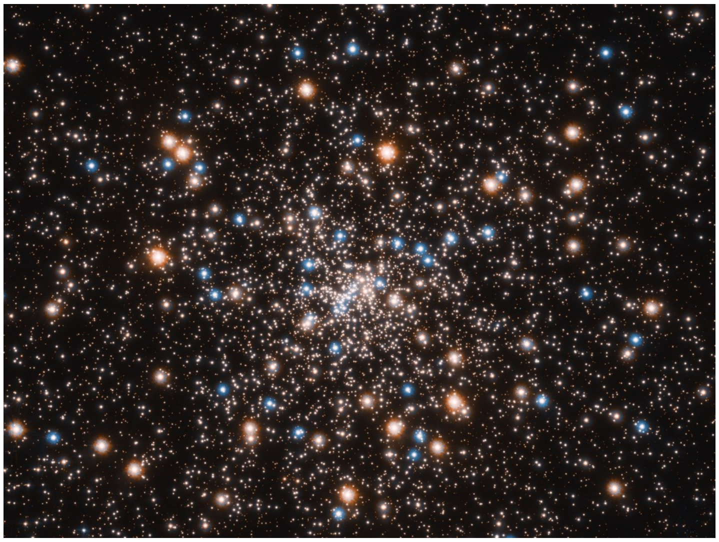
Subsequent NASA missions, in cooperation with ESA, will send spacecraft to Mars to collect these cached samples from the surface and return them to Earth for in-depth analysis.

This is the first high-resolution, color image to be sent back by the Hazard Cameras (Hazcams) on the underside of NASA's Perseverance Mars rover after its landing on Feb. 18, 2021.

Credits: NASA

Hubble Uncovers Concentration of Small Black Holes

Credit: NASA



The amount of mass a black hole can pack away varies widely from less than twice the mass of our Sun to over a billion times our Sun's mass. Midway between are intermediate-mass black holes weighing roughly hundreds to tens of thousands of solar masses. So, black holes come small, medium, and large.

Credits: NASA, ESA, T. Brown, S. Casertano, and J. Anderson (STScI)

Astronomers found something they weren't expecting at the heart of the globular cluster NGC 6397: a concentration of smaller black holes lurking there instead of one massive black hole.

Globular clusters are extremely dense stellar systems, which host stars that are closely packed together. These systems are also typically very old — the globular cluster at the focus of this study, NGC 6397, is almost as old as the universe itself. This cluster resides 7,800 light-years away, making it one of the closest globular clusters to Earth. Due to its very dense nucleus, it is known as a core-collapsed cluster.

At first, astronomers thought the globular cluster hosted an intermediate-mass black hole. These are the long-sought "missing link" between supermassive black holes (many millions of times our Sun's mass) that lie at the cores of galaxies, and stellar-mass black holes (a few times our Sun's mass) that form following the collapse of a single massive star. Their mere existence is hotly debated. Only a few candidates have been identified to date

"We found very strong evidence for an invisible mass in the dense core of the globular cluster, but we were surprised to find that this extra mass is not 'point-like' (that would be expected for a solitary massive black hole) but extended to a few percent of the size of the cluster," said Eduardo Vitral of the Paris Institute of Astrophysics (IAP) in Paris, France.

To detect the elusive hidden mass, Vitral and Gary Mamon, also of IAP, used the velocities of stars in the cluster to determine the distribution of its total mass, that is the mass in the visible stars, as well as in faint stars and black holes. The more mass at some location, the faster the stars travel around it.

The researchers used previous estimates of the stars' tiny proper motions (their apparent motions on the sky), which allow for determining their true velocities within the cluster. These precise measurements for stars in the cluster's core could only be made with Hubble over several years of observation. The Hubble data were added to well-calibrated proper motion measurements provided by the European Space Agency's Gaia space observatory which are less precise than Hubble's observations in the core.

"Our analysis indicated that the orbits of the stars are close to random throughout the globular cluster, rather than systematically circular or very elongated," explained Mamon. These moderate-elongation orbital shapes constrain what the inner mass must be.

The researchers conclude that the invisible component can only be made of the remnants of massive stars (white dwarfs, neutron stars, and black holes) given its mass, extent, and location. These stellar corpses progressively sank to the cluster's center after gravitational interactions with nearby less massive stars. This game of stellar pinball is called "dynamical friction," where, through an exchange of momentum, heavier stars are segregated in the cluster's core and lower-mass stars migrate to the cluster's periphery.

"We used the theory of stellar evolution to conclude that most of the extra mass we found was in the form of black holes," said Mamon. Two other recent studies had also proposed that stellar remnants, in particular, stellar-mass black holes, could populate the inner regions of globular clusters. "Ours is the first study to provide both the mass and the extent of what appears to be a collection of mostly black holes in the center of a core-collapsed globular cluster," added Vitral.

The astronomers also note that this discovery raises the possibility that mergers of these tightly packed black holes in globular clusters may be an important source of gravitational waves, ripples through spacetime. Such phenomena could be detected by the Laser Interferometer Gravitational-Wave Observatory experiment, which is funded by the National Science Foundation and operated by Caltech in Pasadena, California and MIT in Cambridge, Massachusetts.

SJAA Member Astrophoto Gallery

Date: February 10, 2021

Location: Palo Alto, CA

Imaging Telescope: Celestron EdgeHD 11 C11HD

Imaging Camera: ZWO ASI1600MM-Cool Pro

Mount: iOptron CEM60 cem60

Software: My Software PHD2, PixInsight, Lightroom 6.4, Sequence Generator Pro, Photoshop CS6, StarNet++, Topaz Lab DeNoiseAI

SJAA Member Astrophoto Gallery

Dates: Dec 27,2020

Location: Stellar Skies Observatory, Pontotoc, Tx, United States

Telescope: Orion 10" f/3.9 Astrograph

Camera: QSI 683 wsg-8

Mount: Software Bisque Paramount Paramount MYT

Software: Software Bisque Sky X Pro, PixInsight 1.8 PixInsight, Seqence Generator Pro, PHD2 Guiding

SJAA Member Astrophoto Gallery

Date: February 6, 2021

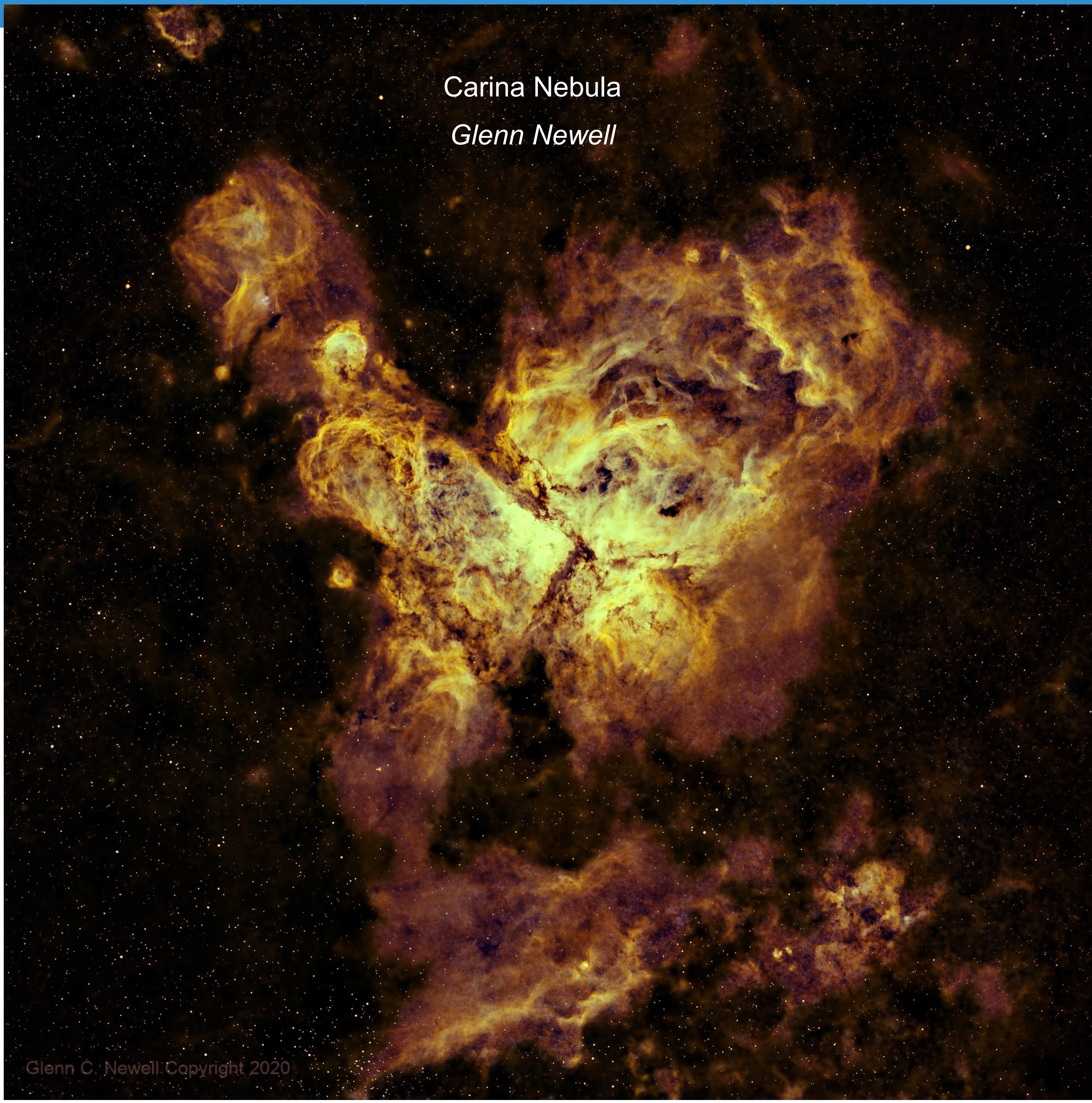
Location: San Jose, CA

Telescope: Takahashi FS 60CB Takahashi fs60cb

Camera: Canon T5i

Mount: Orion Sirius EQ-G Mount Orion Sirius EQ-G Mount

Software: AstroPixelProcessor , StarTools 1.7 Star Tools, Photoshop CC 2018 Photoshop CC2018 , BackyardE-OS, Adobe Lightroom CC



Date: January 5, 2021

Location: El Sauce Observatory, Río Hurtado, Coquimbo Region, Chile

Telescope: Planewave CDK24

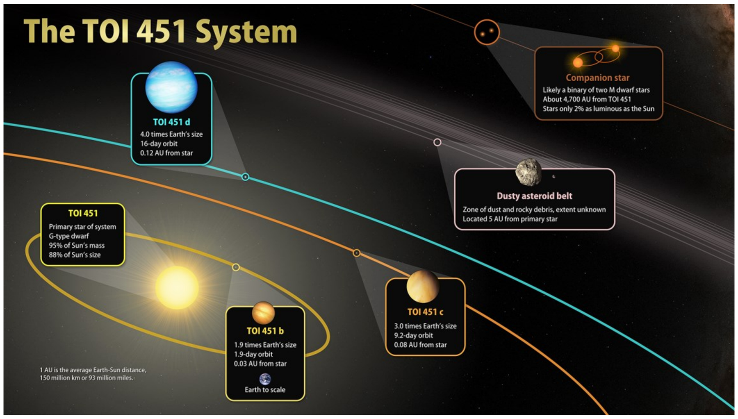
Camera: FLI-PL09000

Mount: Mathis Instruments MI-1000/1250

Software: PixInsight PI 1.8 , Adobe PS CC

NASA's TESS Discovers New Worlds in a River of Young Stars

Credit TESS



This illustration sketches out the main features of TOI 451, a triple-planet system located 400 light-years away in the constellation Eridanus.

Credit: NASA’s Goddard Space Flight Center

Using observations from NASA’s Transiting Exoplanet Survey Satellite (TESS), an international team of astronomers has discovered a trio of hot worlds larger than Earth orbiting a much younger version of our Sun called TOI 451. The system resides in the recently discovered Pisces-Eridanus stream, a collection of stars less than 3% the age of our solar system that stretches across one-third of the sky.

The planets were discovered in TESS images taken between October and December 2018. Follow-up studies of TOI 451 and its planets included observations made in 2019 and 2020 using NASA’s Spitzer Space Telescope, which has since been retired, as well as many ground-based facilities. Archival infrared data from NASA’s Near-Earth Object Wide-Field Infrared Survey Explorer (NEOWISE) satellite – collected between 2009 and 2011 under its previous moniker, WISE – suggests the system retains a cool disk of dust and rocky debris. Other observations show that TOI 451 likely has two distant stellar companions circling each other far beyond the planets.

“This system checks a lot of boxes for astronomers,” said Elisabeth Newton, an assistant professor of physics and astronomy at Dartmouth College in Hanover, New Hampshire, who led the research. “It’s only 120 million years old and just 400 light-years away, allowing detailed observations of this young planetary system. And because there are three planets between two and four times Earth’s size, they make especially promising targets for testing theories about how planetary atmospheres evolve.”

A paper reporting the findings was published on Jan. 14 in The Astronomical Journal and is available online.

Stellar streams form when the gravity of our Milky Way galaxy tears apart star clusters or dwarf galaxies. The individual stars move out along the cluster’s original orbit, forming an elongated group that gradually disperses.

In 2019, a team led by Stefan Meingast at the University of Vienna used data from the European Space Agency’s Gaia mission to discover the Pisces-Eridanus stream, named for the constellations containing the greatest concentrations of stars. Stretching across 14 constellations, the stream is about 1,300 light-years long. However, the age initially determined for the stream was much older than we now think.

Later in 2019, researchers led by Jason Curtis at Columbia University in New York City analyzed TESS data for dozens of stream members. Younger stars spin faster than their older counterparts do, and they also tend to have prominent starspots – darker, cooler regions like sunspots. As these spots rotate in and out of our view, they can produce slight variations in a star’s brightness that TESS can measure.

The TESS measurements revealed overwhelming evidence of starspots and rapid rotation among the stream’s stars. Based on this result, Curtis and his colleagues found that the stream was only 120 million years old – similar to the famous Pleiades cluster and eight times younger than previous estimates. The mass, youth, and proximity of the Pisces-Eridanus stream make it an exciting fundamental laboratory for studying star and planet formation and evolution.

“Thanks to TESS’s nearly all-sky coverage, measurements that could support a search for planets orbiting members of this stream were already available to us when the stream was identified,” said Jessie Christiansen, a co-author of the paper and the deputy science lead at the NASA Exoplanet Archive, a facility for researching worlds beyond our solar system managed by Caltech in Pasadena, California. “TESS data will continue to allow us to push the limits of what we know about exoplanets and their systems for years to come.”

The young star TOI 451, better known to astronomers as CD-38 1467, lies about 400 light-years away in the constellation Eridanus. It has 95% of our Sun’s mass, but it is 12% smaller, slightly cooler, and emits 35% less energy. TOI 451 rotates every 5.1 days, which is more than five times faster than the Sun.

TESS spots new worlds by looking for transits, the slight, regular dimmings that occur when a planet passes in front of its star from our perspective. Transits from all three planets are evident in the TESS data. Newton’s team obtained measurements from Spitzer that supported the TESS findings and helped to rule out possible alternative explanations. Additional follow-up observations came from Las Cumbres Observatory – a global telescope network headquartered in Goleta, California – and the Perth Exoplanet Survey Telescope in Australia.

Kid Spot

Kid Spot Jokes:

- **What is a light year?**
  Same as a regular year but with less calories.
- **How did the astronaut serve drinks?**
  In Sun glasses
- **How does the Solar system hold up its pants?**
  With the Asteroid belt
- **When do astronauts have lunch?**
  At launch time

Goddard's Core Flight Software Chosen for NASA's Lunar Gateway

Credit: NASA

This illustration shows the Gateway lunar platform orbiting the Moon.

Credits: NASA

NASA is improving a flight software system to help create and certify essential software for the lunar Gateway

As part of the Artemis program, NASA will send astronauts to the Moon and establish a sustained lunar presence by the end of the decade. The Gateway will provide a waypoint for lunar exploration and allow astronauts to live and work in lunar orbit as well as host science instruments and experiments.

While Gateway will not be continuously inhabited like the International Space Station, every system onboard must be at a high standard that guarantees astronaut safety. Class A certification assures that all of Gateway's systems meet these rigorous requirements.

NASA, industry partners, and international space agencies are working together to develop Gateway. Goddard Space Flight Center in Greenbelt, Maryland, is collaborating with NASA's Johnson Space Center in Houston to Class A certify the core Flight System (cFS).

The cFS will be essential to Gateway's day-to-day operations, and provides the foundation for Gateway flight software, including the Vehicle System Manager, which manages spacecraft instruments and systems while maintaining core functions.

Gateway's software builds on cFS's dynamic development environment and component-based, adaptable design. Its flexible, layered architecture allows engineers to rapidly assemble significant portions of a software system for new missions. This results in cost and time savings, as mission teams can avoid developing brand new software for each mission.

Conceived in 2004, the open-sourced cFS software has been improved both internally and through recommendations from independent developers around the world. "We're working on making it easier to test, easier to trace requirements from mission applications, and easy to adapt," said Jacob Hageman, team lead for the ongoing certification effort for Gateway's cFS. "The Artemis program provides resources to help us improve the product, which benefits everyone who uses it."

Goddard developers envisioned an independent, reusable software framework for routine spacecraft tasks, including telemetry, health and safety, and stored commanding. In 2008, the Lunar Reconnaissance Orbiter launched, operating on the core Flight Executive – a plug-and-play foundation for what would become cFS.

Goddard flight software architect Jonathan Wilmot has worked on cFS from its beginning, when the idea was born out of a need for efficiency. "We had two big missions at Goddard at one time, the Solar Dynamics Observatory and the Global Precipitation Measurement," he said. "There wasn't enough staff to do both independently, so we worked with Goddard's software and mission teams to establish a set of requirements."

From the Board of Directors

Announcements

◊ San Jose Astronomical Association is monitoring closely the coronavirus, COVID-19, situation in Santa Clara county. In accordance with advisement from Santa Clara County Health Department we may be canceling some events in the next few weeks, as appropriate - especially those that involve close contact and shared surface on telescopes. Please check our Meetup to determine if an event has been canceled before heading out to the event. The board will periodically review the status of the outbreak and the guidance from the Santa Clara county to determine the impact upon upcoming events.

◊ Ken Miura, Vini Carter, Swami Nigam, Peter Melhus and Sukhada Palav were reelected into their respective Board of Director positions.

◊ New officer elections for President, Vice-President, Treasurer, and Secretary positions will be held online via electionbuddy between March 25th and 27th and the new officers will be announced at the March board meeting. Any member interested in running for these positions should reach out to the Secretary at secretary@sjaa.net for eligibility criteria and other details.

Board Meeting Excerpts

January 2, 2021

In attendance
Swami, Sandy, Glenn, Ken, Emi, Peter, Wolf, Vini Sukhada, Rob
Excused Absences: None
Guests: Beth, Kanch, Rishabh, Marianne

Board Meeting Logistics and Efficacy
Ken proposes to extend the monthly board member meeting duration from the current 1.5hours to ensure a more effective use of the meeting time and cover most of the topics on the agenda.

Improve SJAA member communication
Kanch proposes to send a monthly communication email after the board meeting to coordinate with Beth and Ephemeris editors.

Online voting for board and officer election
The voting for the 2021 SJAA board and officer positions will be done online via electionbuddy. The board seats up for election in February 2021 are Directors 1, 3, 5, 7, and 9 (incumbents: Vini, Ken, Swami, Peter, and Sukhada).

January 30, 2021

In attendance
Swami, Emi, Vini, Glenn, Sukhada, Wolf , Vini, Rob J, Ken, Peter
Unexcused Absences: Sandy
Guests: Dave, Marianne, John Gates

Upcoming Board Election
Board elections are coming up in February and 5 of the 9 board positions are up for elections.

Bob Ayers' Estate
Information regarding Bob Ayers' donations to SJAA are on page 3 of this issue.

24" Dob Telescope
Five possible options are identified for utilization of the 24" Dob. Action is being taken by the task force group and an update will be provided soon.

Virtual Outreach request from FAAC
SJAA has received an inquiry from a past member (Sandra Macika) who is affiliated with another astronomy club (Ford Amateur Astronomy Club) in Michigan. She would like SJAA to make a virtual presentation on astronomy to her club. A person from the club is yet to be assigned to deliver talks.

February 27, 2021

In attendance
Wolf, Vini, Rob J, Glenn, Sandy, Ken, Swami, Emi
Excused Absences:
Guests: Marianne, Kanch

Board member Elections
Board member election results for Director 1, Director 3, Director 5, Director 7 and Director 9 were announced during the February 27 board meeting. All incumbents - Vini Carter, Ken Miura, Swami Nigam, Peter Melhus, Sukhada Palav were reelected.

Continued use of the hall at Houge Park
SJAA submitted an RFQ with the City of San Jose's (CSJ) Parks Department for continued use of the Hall at Houge Park. The board awaits updates from CSJ.

Discontinuing of DSL Internet Connection at Houge Park
Sukhada and Rob have been notified by ATT that costs will rise unless a physical upgrade is made to the site. An executive action was made to discontinue the service until Houge Park is allowed to be open once again.

Officer Elections
Officer elections for President, Vice-President, Treasurer, and Secretary positions will be held online via electionbuddy between March 25th and 27th and the new officers will be announced at the March board meeting. Any SJAA member in good standing that is interested in running for an officer seat is welcome, and they should email the Secretary by 20th of March. Officer duties are described in Article 4 of Club By-Laws.

https://www.sjaa.net/about/club-by-laws/#A4

SJAA Library

SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

http://www.sjaa.net/programs/fixit/

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

http://www.meetup.com/SJ-Astronomy/

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

http://www.sjaa.net/programs/beginners-astronomy/

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.

http://www.sjaa.net/programs/loaner-telescope-program/

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

http://www.sjaa.net/programs/imaging-sig/

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for additional information.

http://www.sjaa.net/programs/school-star-party/

SJAA Contacts

President/Dir:	Ken Miura
Vice President:	Vacant
Treasurer/Dir:	Rob Jaworski
Secretary:	Emi Nikolov
Director:	Vini Carter
Director:	Sandy Mohan
Director:	Wolf Witt
Director:	Swami Nigam
Director:	Peter Melhus
Director:	Glenn Newell
Director:	Sukhada Palav
Ephemeris Newsletter -	
Content Editor:	Abishek Ramasubramanian
Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
In-Town Star Party:	Muditha Kanchana
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal Gangakhedkar
Publicity:	Rob Jaworski
Donations:	Dave Ittner
Solar:	Wolf Witt
Starry Nights:	Carl Wong
School Events:	Cat Cvengros
Refreshments:	Marianne Damon
Social Media Manager:	Beth Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki

E-mails: http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
http://www.sjaa.net/contact

Ephemeris Printing Questionnaire

Please help us the Ephemeris production team by providing your input to the following questions. You may do so by either going to this URL: https://tinyurl.com/yxaqjhmq or by filling it in and mailing it to: SJAA, P.O. Box 28243 San Jose, CA 95159-8243.

1. Your Name (Optional):

2. In the last few editions, we moved the Astronomy pictures to the center 4 pages and made them colored. What do you think of this change:

a)Love it

b)Hate it

c)Didn't care or I don't get a printed copy

d)Didn't even notice the change

3. We have explored the cost of printing Ephemeris in 3 possible formats. Color printing significantly increases the cost, so it will require us to increase the annual amount we charge for printed newsletter. Note, the full color PDF version is always available on our website. Please indicate which of the following 3 options do you prefer:

a)Full B&W printing - member cost stays at $10 per year

b)Print 4 pages with astroimages in color, other pages B&W - member cost increases from $10 to $15 per year

c)Print all pages in color - member cost increases from $10 to $20 per year

4. If there are any other suggestions regarding Ephemeris that you would like to share with us, please write them here:

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243

Place
postage
here

Fold here

San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

☐ **New** ☐ **Renewal**

☐ **$20** Regular Membership with **online** Ephemeris

☐ **$30** Regular Membership with **hardcopy** Ephemeris mailed to below address

The newsletter is always available online at: http://www.sjaa.net/sjaa-newsletter-ephemeris/

Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: http://www.sjaa.net/join-the-sjaa/

Name: ____________________

Address: ____________________

City/ST/Zip: ____________________

Phone: ____________________

E-mail address: ____________________